

Reserve review: US commercial auto (NAIC 1767) as at 31 December 2007

Illustrative reserving study on NAIC Schedule P data

Data: CAS Loss Reserving Database (NAIC Schedule P), compiled by S&P Global Market Intelligence for the Casualty Actuarial Society. Net of reinsurance, US business. The IFRS 17 and Solvency II figures are illustrative methodology demonstrations, not accounting or regulatory numbers for any company. Insurers are identified by NAIC code only. Amounts in USD millions unless stated.

1. Executive reserve review

Selected unpaid claims at 31 December 2007 are USD 365.4m, net of reinsurance, for accident years 1998–2007. This is USD 112.3m (44%) above the insurer's own booked unpaid of USD 253.1m. The estimate covers commercial auto losses including defence and cost containment (DCC), undiscounted and excluding ULAE. It was locked in git (tag `selection-locked`) before any later development was looked at.

	USD m
Selected ultimate, AY 1998–2007	1876.9
Paid to date	1511.5
Selected unpaid	365.4
of which case reserves	117.0
of which IBNR (incl. IBNER)	248.4
Insurer's booked unpaid (case + bulk/IBNR)	253.1
Range across methods (paid CL to incurred BF)	343.5 – 370.2

Main drivers.

- AY 2006–2007 account for USD 255.4m (70%) of the unpaid. They are 58% and 33% paid-developed, and are selected with Benktander at loss ratios of 61.9% and 63.8%.
- The insurer books AY 2007 at a 49.0% loss ratio and AY 2006 at 53.7%. That implies much faster payment than its own history supports: paid chain ladder gives 61.4% for AY 2007.
- The insurer's own first bookings have always developed upwards: every AY 1998–2006 booked ultimate has risen since its first valuation, by 8% to 30%. This pre-lock evidence supports holding more than the booked figure.
- The tail beyond 120 months is 1.0023 (log-linear extrapolation). It is immaterial, at under 0.3% of ultimate.

2. Evidence and uncertainty

Data and diagnostics. The paid and case-incurred triangles are complete and clean: no negative paid increments and no negative case reserves. The paid ÷ incurred ratio is stable by accident year, so neither faster settlement nor a change in case-reserve strength is evident. Mid-lag paid factors have risen for recent accident years (the 3–4 factor averages 1.13 for AY 1998–2001 and 1.16–1.18 for AY 2002–2004), and incurred factors show the same rise. `cyEffTest` does not reject (paid $Z = 12$), while `dfCorTest` rejects independence for paid, so Mack standard errors are treated as a benchmark only. Factors are volume-weighted over the latest five diagonals.

Pseudo-historical check (pre-lock). The rules were rerun at 2004, 2005 and 2006 and compared with paid diagonals up to 2007. Paid chain ladder under-predicted payments every time (A/E 1.15, 1.20, 1.09). The default rule did better (1.08, 1.14, 1.07). Using incurred chain ladder for mature years brought the A/E closer to 1 (`outputs/m4_pseudo_historical_by_ay.csv`). The latest-5 and all-year windows gave the same A/E to within 0.5%.

Selection by accident year. Default rule on paid percent developed z : chain ladder at $z \geq 70\%$, Benktander at 30–70%, and BF below 30%. BF uses a leave-one-out Cape Cod prior, 65.3%–67.2%. One documented exception applies: for mature years (AY 1998–2005, $z \geq 70\%$), incurred chain ladder replaces paid chain ladder, based on the pseudo-historical evidence above. No accident year shows a paid vs incurred chain ladder gap above 10%. The prior is more than 10 points above the peer median loss ratio for recent years, but the peer panel is mostly small insurers with a different mix, so the prior was kept and its ± 5 -point sensitivity is part of the uncertainty analysis.

Table 1: Ultimates by method and selection, USD m (z = paid percent developed; CL_INC = incurred chain ladder; BK = Benktander).

AY	Prem	Paid	z	Paid CL	Inc CL	Paid BF	Benkt.	Method	Selected	ULR	Unpaid
1998	245.0	158.0	100%	158.4	158.4	158.4	158.4	CL_INC	158.4	64.6%	0.4
1999	231.5	158.4	100%	159.2	159.1	159.1	159.2	CL_INC	159.1	68.7%	0.7
2000	222.2	170.7	99%	172.7	175.8	172.4	172.7	CL_INC	175.8	79.1%	5.1
2001	233.6	149.9	98%	153.1	152.4	153.1	153.1	CL_INC	152.4	65.3%	2.5
2002	259.3	173.8	97%	180.0	180.3	179.7	180.0	CL_INC	180.3	69.5%	6.5
2003	281.5	173.3	94%	185.0	187.1	185.1	185.0	CL_INC	187.1	66.5%	13.8
2004	301.6	159.2	87%	182.3	182.8	184.9	182.6	CL_INC	182.8	60.6%	23.7
2005	322.8	167.3	76%	220.9	224.6	219.2	220.5	CL_INC	224.6	69.6%	57.3
2006	354.9	126.1	58%	216.0	218.9	225.1	219.8	BK	219.8	61.9%	93.7
2007	370.6	74.7	33%	227.6	227.5	240.9	236.5	BK	236.5	63.8%	161.8

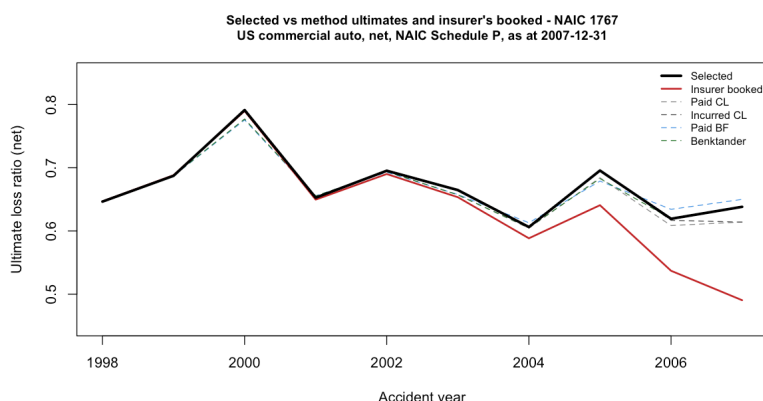


Figure 1: Selected vs method ultimates and the insurer's booked ultimate (loss ratio). US commercial auto, net, NAIC Schedule P, as at 2007-12-31.

Uncertainty. All distributions use the stochastic specification frozen at the lock. The custom ODP bootstrap resamples the selected reserve itself. On each of 10,000 pseudo-triangles it re-estimates the factors, the tail and the leave-one-out prior (with 10% lognormal prior noise), then applies the locked method table. The incurred-CL uplift for AY 1998–2005 is carried as a fixed load run off with the paid pattern.

Model	Mean	SD	CV	75th pct	99.5th pct
Mack, paid (with tail)	340.2	19.2	5.6%	–	–
Mack, incurred (with tail)	347.4	18.6	5.4%	–	–
Package ODP bootstrap (pure CL, no tail)	335.9	18.5	5.5%	348.0	386.1
Custom bootstrap of selected reserve	363.0	22.8	6.3%	377.9	426.8

Mack splits the paid SE into process 15.4m and parameter 11.5m. The custom bootstrap is wider than the pure-CL models (CV 6.3%) because it adds prior uncertainty and the BF weight on young years. Judgemental assumptions outside the bootstrap: +2% a year calendar inflation adds 4.6%; ELR ± 5 points moves unpaid by $\pm 3.1\%$; incurred instead of paid lowers it by 1.4%; tail $\times 0.5/\times 1.5$, window and threshold changes move it by 1.2% or less ([outputs/m6_sensitivities.csv](#)).

Application layer (illustrative). Discounting the simulated cash flows (plus 5% ULAE) on the GSW US Treasury zero curve at 31 December 2007 gives a discounted best estimate of USD 360.6m. A 75% confidence-level risk adjustment of USD 14.7m (4.1% of BE) gives an **illustrative net IFRS 17 LIC of USD 375.2m**. The risk adjustment corresponds to a 75% confidence level. Re-reserving 10,000 simulated 2008 years with the locked rules gives a 1-in-200 one-year deterioration of **12.0% of opening BE**, against 27% ($3 \times 9\%$) under the Solvency II standard formula for reserve risk. The Merz–Wüthrich one-year SE for pure chain ladder is USD 14.4m. This is a reserve-risk-only benchmark on US data, not an SCR.

3. Validation and limitations (post-reveal addendum)

Written after the 2008–2016 run-off was revealed. Page 1 and the first half of page 2 were fixed before the reveal and are unchanged; the uncertainty, LIC and one-year figures on page 2 were computed before the reveal.

Blind back-test. Scored with the formulas fixed in [CONTRACT.md](#), reading the selection from the tag:

- **Test A (paid to 120 months, headline).** Payments projected for 2008–2016 came to USD 360.3m against USD 401.7m actually paid. The locked estimate was **10.3% short**; pure paid chain ladder was 15.6% short.
- **Test B (120-month incurred proxy, AY 1999–2007).** Against the proxy unpaid of USD 411.3m, the selection was 11.2% low, pure chain ladder 16.6% low and the insurer's booked reserve 38.6% low. **The locked selection beat both pure chain ladder and the insurer's own booked estimate.**
- **Test C (2008 A/E).** Actual 2008 payments were $1.01 \times$ expected, so the first year was on track.
- **Test D (rolling re-reserve).** Mechanical re-reserving showed deterioration in 2008 (USD 10.1m), 2009 (11.0m) and 2010 (19.3m), with small movements after that. The 2008 deterioration sits at the 26.9% point of the simulated one-year distribution, well inside it.

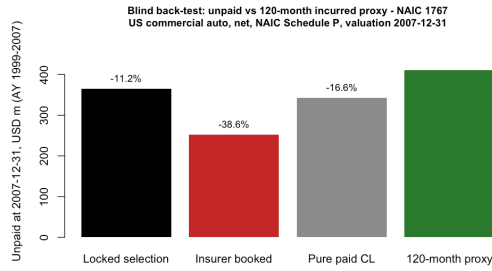


Figure 1: Unpaid at 2007 vs the 120-month incurred proxy (AY 1999–2007)

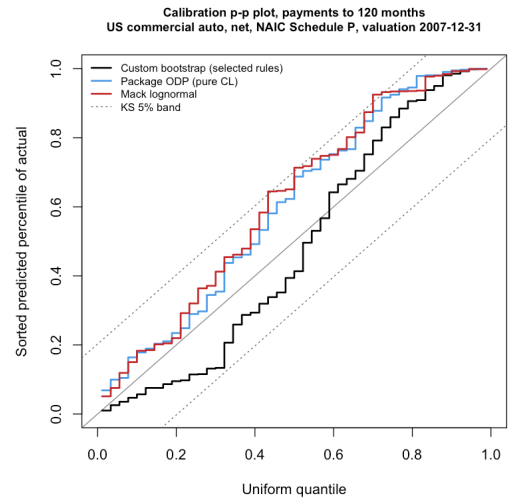


Figure 2: Calibration p-p plot across panel insurers

US commercial auto, net, NAIC Schedule P, valuation 2007-12-31.

What hindsight would change, and was it knowable? Late development after 2007 exceeded every method. The pre-lock evidence did point this way: heavier recent mid-lag factors, pseudo-historical A/E above 1 in every test, and an insurer whose own bookings had always developed upwards. Those signals moved the selection above chain ladder and far above the booked figure, but not far enough. With hindsight, an explicit calendar-year trend, or stronger weight on incurred for AY 2004–2007, would have closed more of the gap; the +2% inflation sensitivity alone covered less than half of it.

Across insurers. The same automatic rules on 47 insurers reduced the median absolute Test A error from 24.5% (pure chain ladder) to 20.8%. The gain is not uniform: the rule beat chain ladder for about half of the insurers, and the smallest third have errors above 50%. Only 19.1% of insurers had booked 2007 reserves below the proxy, so this insurer’s under-reserving was unusual.

Calibration. On 45 insurers, the custom bootstrap’s 75th percentile covered 66.7% of actual outcomes (nominal 75%). The two-sided 50% and 90% intervals covered 33.3% and 77.8%, and the p-p plot is S-shaped, so the intervals are too narrow rather than off-centre. KS D = 0.18 against a critical value of 0.20 (not rejected at 5%). The package ODP and Mack distributions covered only 57.8%, sit above the diagonal (biased low) and are rejected by KS. This is consistent with Meyers (2019) on the 1988–97 data. A widening factor fitted on half the panel ($\lambda = 1.40$) cut the two-sided coverage gap on the other half from 0.53 to 0.10. This is reported as a sensitivity only. The one-year view was better calibrated: the one-year 75th percentile covered 71.1% of actual 2008 CDRs.

Limitations. US business in USD at 2007. Net of reinsurance, so there is no gross or reinsurance-held IFRS 17 measurement. The data has no claim counts, ULAE or rate changes, so frequency–severity, Berquist–Sherman and on-levelling are not possible and the BF prior is not on-levelled. The back-test runs only to 120 months, so the tail is untested. One vintage (AY 1998–2007) and small panels (47 / 45 insurers) mean calibration results are indicative and not extrapolated. The CAS selection of insurers with complete data introduces survivorship bias. IFRS 17 did not exist in 2007, so the LIC is a method demonstration. The one-year measure covers reserve risk only and is not an SCR. The 10% prior CV and 5% ULAE are judgements.